

EDUCATION

Purdue University · B.S. Computer Engineering · Cumulative GPA 3.5 West Lafayette, IN · May 2028

EXPERIENCE

Undergraduate Researcher, Computational Quantum Electromagnetics Lab Aug – Dec 2025

Under Dr. Roth, Purdue University

- Engineered a Python automation framework for GMSH to streamline 3D mesh generation, reducing manual Abaqus input file preparation time by 50–70%.
- Developed modular scripts for conformal meshing and physical group assignment, producing FEM-ready structures.
- Authored technical documentation ensuring 100% reproducibility for future large-scale simulations.

Undergraduate Researcher & OUR Scholar Aug 2025 – May 2027 (ongoing)

re3data Recommendation Engine, Under Prof. Witt, Purdue University

- Selected as an OUR Scholar with a \$1,000 research scholarship for AI-enhanced metadata curation of the re3data repository.
- Architected a high-concurrency Flask backend serving content-based recommendations for 3,400+ research repositories, reducing search discovery time by 30%.
- Built semantic search using TF-IDF and cosine similarity, validated via t-SNE clustering.
- Built a reliability-aware API layer with real-time HTTP status checks, increasing accessible repository visibility by 40%.

Teaching Assistant (ILS 103 → ENGR 103), Purdue University Fall 2025 – Present

- Fall 2025 (Engineering in the World of Data):** Graded quizzes, discussion posts, and final projects, resolved student queries, and presented weekly attendance visualizations
- Spring 2026–Present (Applications of AI in Engineering):** Co-designed interactive Colab notebooks on AI/ML concepts (Logistic Regression, XGBoost, Random Forests, CNNs, backpropagation) and led hands-on labs applying ML to real engineering datasets
- Ambassador, Jan–May 2025:** Boosted student engagement within the Learning Community

PROJECTS

Quantum Simulation Engine · C++17/20, numerical methods · Live Demo Jun 2026 – Present

- Built a from-scratch C++ simulation engine — ODE/PDE solvers (RK4, Crank-Nicolson) and a quantum circuit simulator validated via Bell-state and teleportation protocols (1000/1000 trials)
- Engineered a CUDA state-residency optimization, eliminating per-gate transfer overhead for a **95.9× speedup** at 256 gates, verified bit-exact (0.000e+00 error) against CPU
- Applied the roofline model across OpenMP and CUDA kernels — 3.2× compute-bound speedup, diagnosed a transfer-bound GPU regime (96% PCIe overhead) that drove the residency redesign

esh — Unix Shell · C, POSIX Summer 2026

- Built a fully functional Unix shell using only POSIX system calls — fork/exec/wait, pipes, dup2 I/O redirection, and signal handling for Ctrl+C.
- Shipped with zero external frameworks, working directly at the OS internals and hardware-software boundary.

Multi-Client TCP Chat Server · C, socket programming Summer 2026

- Served concurrent clients from a single thread via select-based I/O multiplexing, using the same event-driven architecture as nginx and Redis.
- Implemented stateful sessions, broadcast and private messaging via a command parser, plus a custom TCP client.

Exoplanet Database & Analysis Engine · PostgreSQL, Node.js/Express, Chart.js Summer 2026

- Analyzed 5,700+ NASA-confirmed exoplanets in a full-stack app, using PostgreSQL window functions, CTEs, and aggregations to verify Kepler's third law and compute stellar habitable zones.
- Exposed the analysis through a multi-endpoint REST API with Chart.js frontend visualizations.

SKILLS

Languages	C/C++17/20, Python, SQL, JavaScript (Node.js), SystemVerilog, MATLAB, Bash
Systems	POSIX APIs, socket programming, I/O multiplexing, process management, finite-difference methods, OpenMP, CUDA
Tools	PostgreSQL, Express, Flask, NumPy, Pandas, scikit-learn, Chart.js, Git, Linux/WSL, Valgrind/GDB
Concepts	OS internals, numerical methods, ODE/PDE solvers, GPU memory optimization, FEM mesh generation, statistical modeling
Professional	Cross-functional collaboration, technical communication, instructional mentorship, technical documentation

COMPETITIONS, COURSES & WRITING

- IMC Prosperity 4 — algorithmic trading competition, 2026
- Akuna Capital Options 101 — options theory and derivatives coursework, completed 2026
- The wave equation is a time machine** (Medium, Jun 2026) — original derivation and physical interpretation of D'Alembert's wave equation discretization.